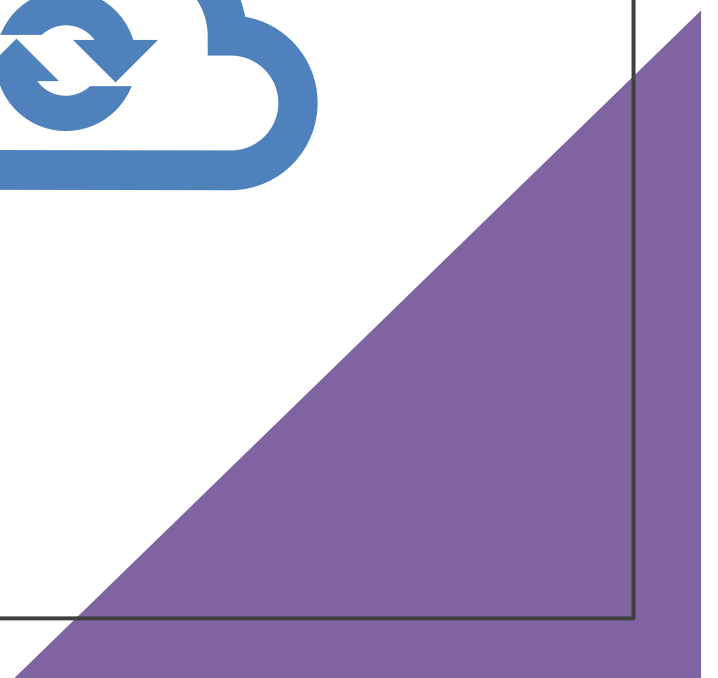
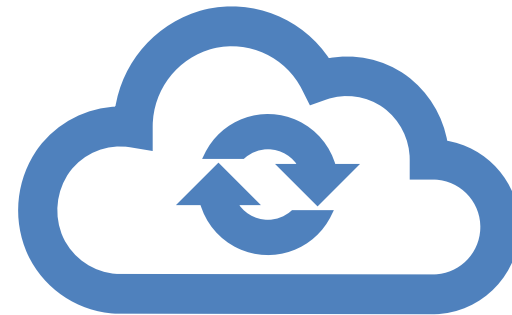


Optimizing CI/CD for Hybrid and Cloud Environments with DevOps and AI

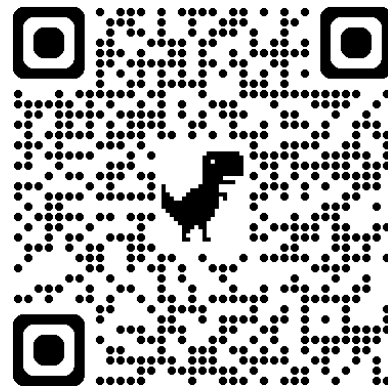
Building scalable, secure, and AI-enhanced CI/CD pipelines



Kellyn Gorman

Multiplatform and AI
Advocate

<https://dbakevlar.com>



Agenda



FOUNDATIONS OF MODERN CI/CD

LEVERAGING AI IN CI/CD PIPELINES

SECURITY AND GOVERNANCE

KEY PRACTICES AND PATTERNS

DEMO AND WALKTHROUGH

Q&A AND TAKEAWAYS

Discussion vs. Demo



Enterprise is EXPECTED



Policy and governance first, solutions and tools later



Cost is what I hear about most often, so...



This is a VIBE CODE – seriously...

Foundations of Modern CI/CD

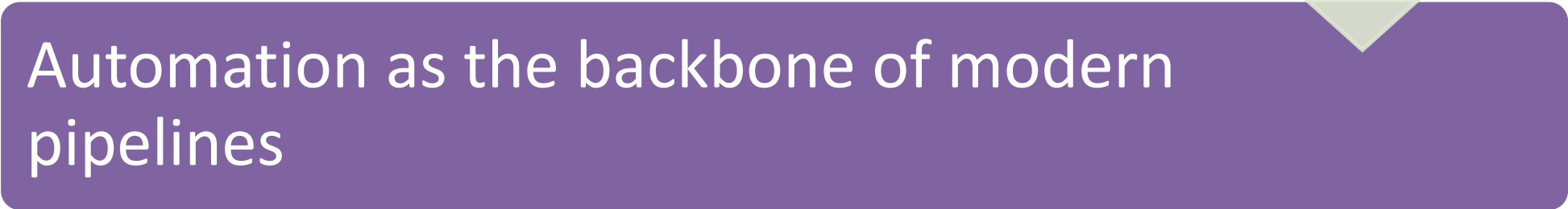
Principles: Continuous Integration and Delivery



Key tools for SCM, CI/CD, Testing, and Observability

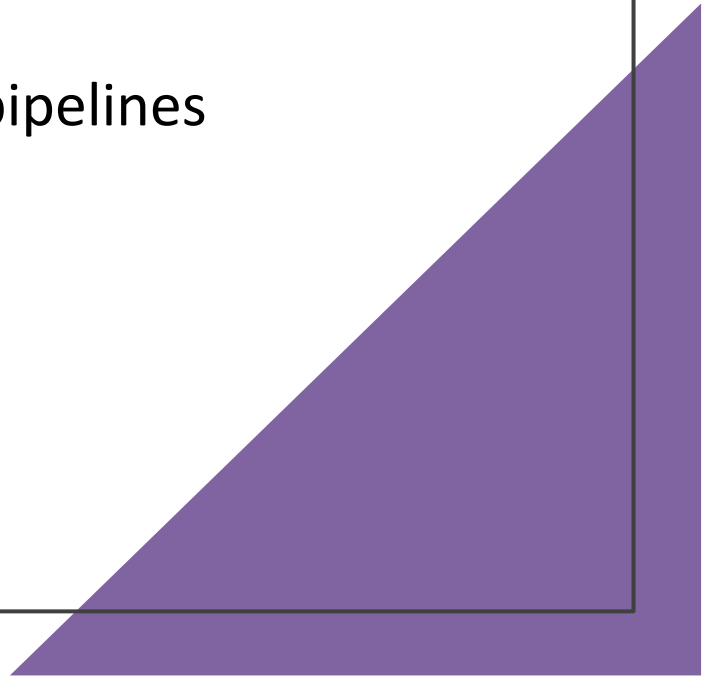


Automation as the backbone of modern pipelines



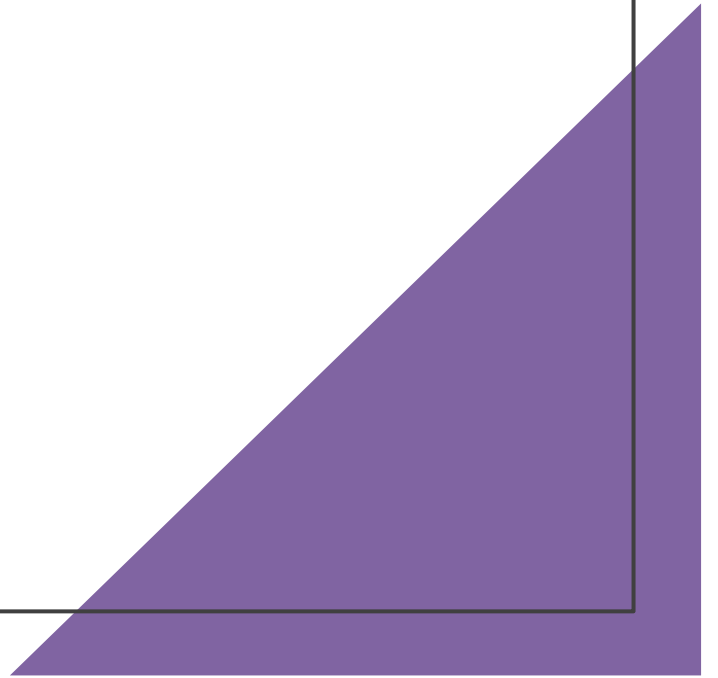
Leveraging AI in CI/CD Pipelines

- Predictive build and test optimization
- AI-assisted code review and recommendations
- Anomaly detection in pipelines and deployments
- Smarter rollbacks and deployment decisions



Security and Governance

- Mitigating AI-related vulnerabilities
- Ensuring model integrity and compliance
- Zero Trust pipelines
- Policy-as-Code and automated compliance checks



Key Practices and Patterns

- GitOps for declarative pipelines
- Infrastructure as Code with Terraform, Pulumi, or Bicep
- Progressive delivery: Canary and blue-green deployments
- Feature flag strategies



The Challenges of Incorporating AI



Capability
Understanding



Goals

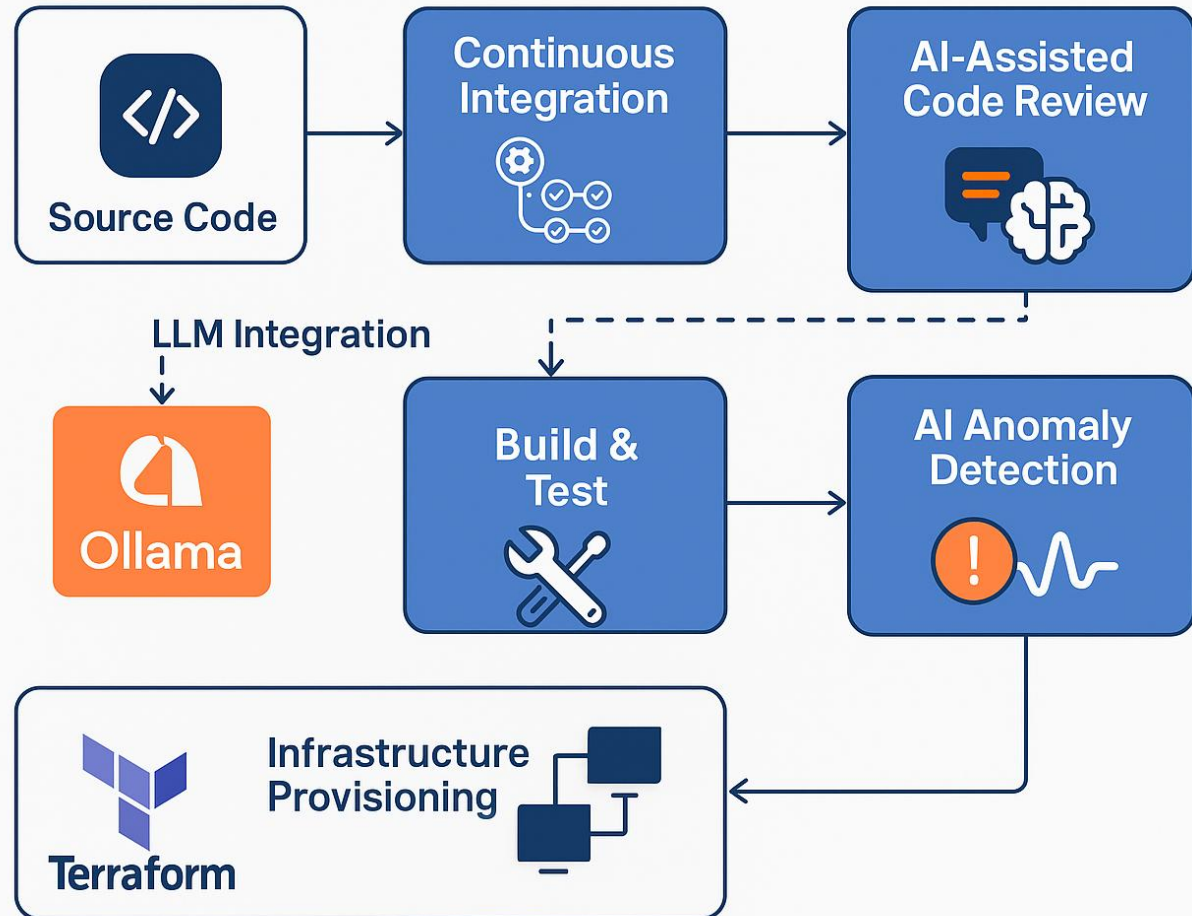


Cost



Trust

Example Workflow





Baseline DB → AI-assisted code review → Automated testing



AI anomaly detection → Canary deployment



No cloud, terraform, OpenAI but could be easily adapted.

Demo Walkthrough

Demo Environment

Ollama on Windows 11 Laptop

Qwen 2.5 LLM

Postgres 17.5 on laptop

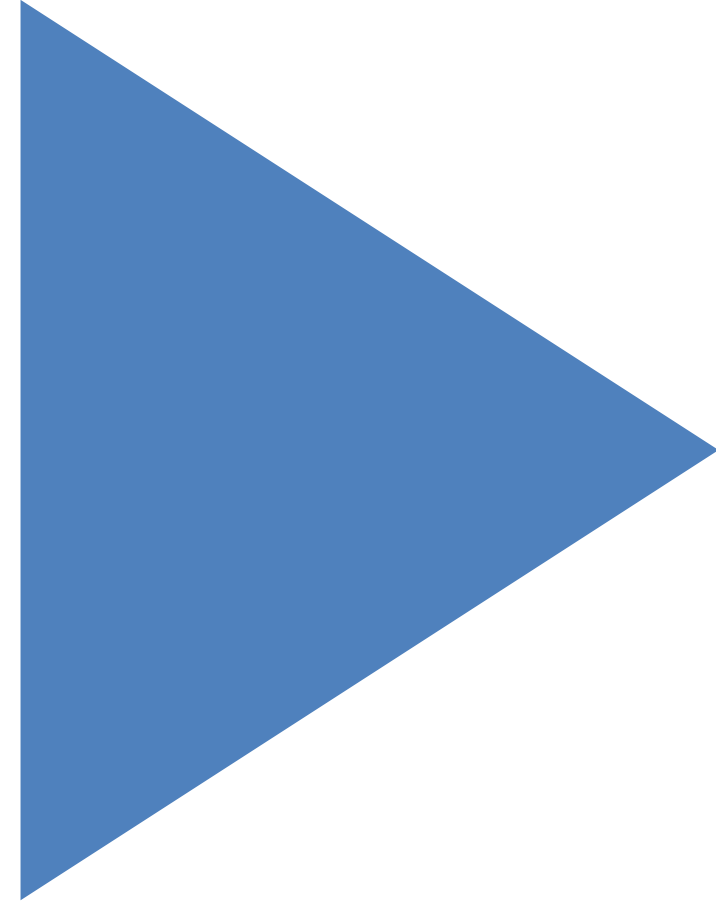
Flyway Teams and CLI

Powershell and SQL scripts

Only CPU laptop- No GPU, tokens, etc.



DEMO



Scripts and Demo Sheet Available at:
<http://github.com/Dbakevlar/Presentations2025/>

Key Takeaways



Automate everything,
with governance baked
in



Balance AI innovation
with security and
compliance



Use observability for
smarter, safer decisions

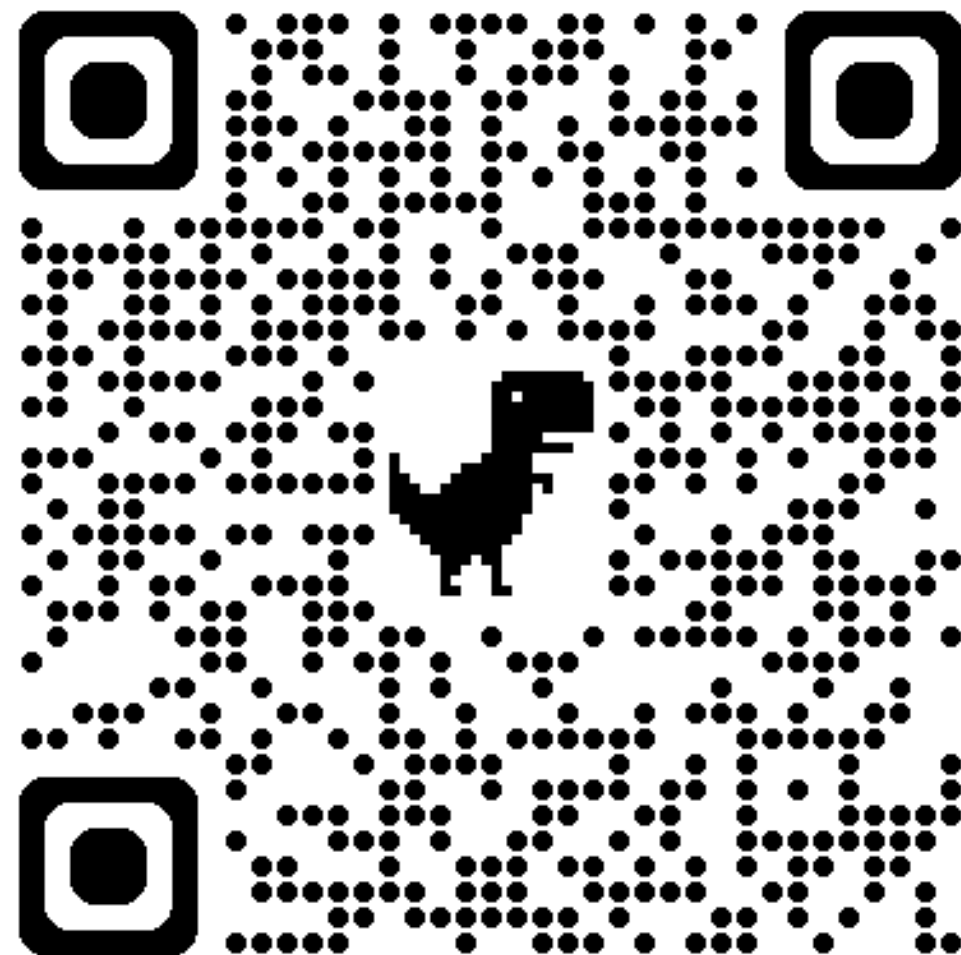


Start small, scale
responsibly



Q&A

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<https://dbakevlar.com>



References

- Ollama for Windows: <https://ollama.com/download/windows>
- Flyway Teams and CLI: <https://www.red-gate.com/hub/product-learning/flyway/automating-flyway-desktop-development>
- Postgres on Windows: <https://www.postgresql.org/download/windows/>
- The qwen LLM (I use 2.5 to keep it very lightweight):
<https://huggingface.co/Qwen>
- All code can be found here:
<https://github.com/Dbakevlar/Presentations2025>